What happens between a sensor reading and a robot's action?



Sensor
reads the world

Hardware
executes the instructions

Network
links devices when needed

Agent
chooses an action

Action
changes the world

Three beginner modules, one traceable system: **Hardware** · **Networks** · **Behaviorist AI**

Who we teach and what beginners will learn

Intended learners

First-year BIT students

No prior course assumed

Enough to start

Everyday computer use
School arithmetic

A module map guides each reading route.

Hardware

CPU and memory
Instruction execution

Learning goal

- Trace an instruction.
- Explain which register changes and why.

Networks

IP and transport
DNS, DHCP, HTTPS, SSH

Learning goal

- Trace a fresh HTTPS request.
- Explain the role of each protocol.

Behaviorist AI

Agents and rewards
Q-learning and DQN

Learning goal

- Calculate a Q-learning update.
- Explain how reward affects an estimate.

Source: CONTENT-OUTLINE.md (outcomes 1-6) and the three module OUTLINE.md maps

Current progress

Hardware

34 intro + 33 lessons in 9 sections

- 179 explained answers
- 35 local SVG figures
- ADD / LOAD traces

Drafted & revised · review pending

Networks

9 chapters, overview to SSH

- 30 explained answers
- 2 local SVG figures
- HTTPS request trace

Drafted & revised · review pending

Behaviorist AI

6 chapters, agents to robotics

- 20 explained answers
- 2 local SVG figures
- Q-learning episode

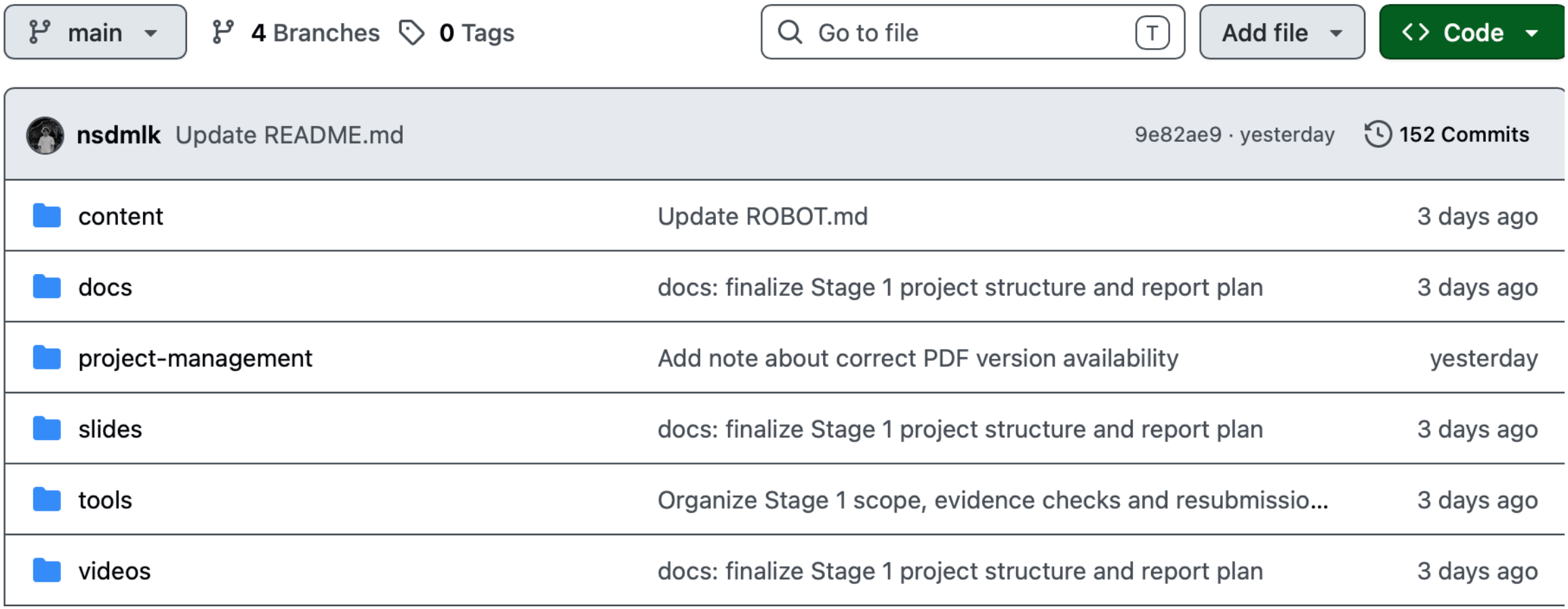
Drafted & revised · review pending

Completed in source	Drafts, worked examples, answer keys, 39 SVG figures, module maps and project records.
Still to finish	Independent content review (A11), syllabus check, rehearsal and offline playback test.
Planned for Stage 2	Study Guide integration and tutorial videos. Stage 2 is due 7 November.

49 learner texts. Source: module maps and TASKS.md. Snapshot: main 9e82ae9, checked 29 October.

Repository organisation

github.com/nsdmlk/bit-ai-and-cs



GitHub main, 9e82ae9. Team screenshot from 29 October. Files are embedded for offline presentation.

A meaningful change: explained answers

26 October 2026 Commit ac5dd0a content/Networks/ANSWER-KEY.md

Earlier draft: questions referred to an answer resource that was still to come.

```
4. **HTTPS trace.** DNS supplies an address for the hostname. The host chooses a
route; in the example, a remote destination uses its default gateway. IP packets
travel through successive links and routers. TCP establishes the connection, then
TLS validates the server identity and establishes protection. The browser exchanges
HTTP requests/responses inside TLS and renders the returned content. Omitting TLS
would omit the protection that makes this example HTTPS.
```

Revision: 30 explained answers, including the missing TLS step.

A learner can now check the reasoning after trying the question.

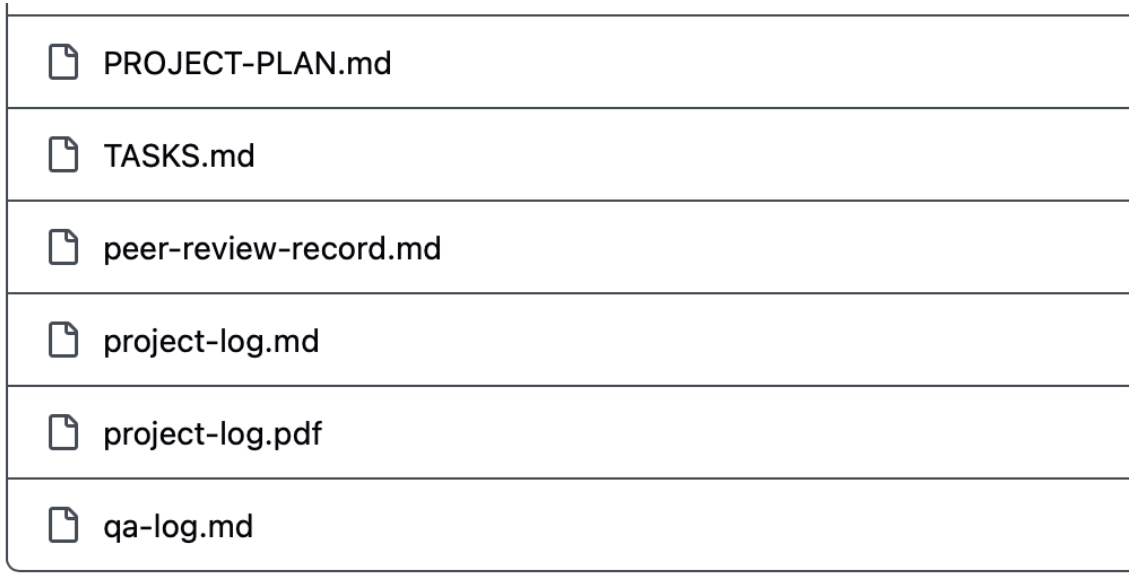
Green lines show a real addition in ac5dd0a. INTRODUCTION.md links its questions to this answer key.

What the team review changed

25 October: team architecture review

We had chapters, but the shared plan and evidence trail were unclear.

The revision added module maps, tasks and a review record.



One important item remains open: independent content review

A11	Complete the independent human check of the revised teaching anchors	Alimzhan	Pending actual review	Check one Hardware, one Networks, and one Behaviourist AI activity; record the real outcome in `project-management/qa-log.md`

Sources: peer-review-record.md and TASKS.md, main 9e82ae9. AI checks support the revision; team content review remains pending.

Stage 2 priorities, due 7 November

1 Independent human review

Alimzhan checks the three teaching anchors and records real findings in qa-log.md.

A11: pending review

2 Scope and Study Guide

Confirm syllabus coverage. Integrate the checked chapters into one Study Guide.

Task A16 · planned

3 Tutorial videos

Develop the module storyboards into tutorials that match the reviewed Markdown.

Stage 2 media: planned

Before class: rehearse under 4:00, test the extracted USB folder, then submit the version used in class. These checks remain to be done.

The actual Markdown example

content/Hardware/03 Fetch Decode Execute/05-worked-examples.md

Example 1: `ADD R1, R2`

Suppose `ADD R1, R2` sits at address `0x100`. PC = `0x100`. R1 = 5, R2 = 3.

Fetch:

- PC = `0x100` → address bus.
- Memory returns the instruction → IR.
- PC becomes `0x104`.

State	Before	After
PC	0x100	**0x104**
IR	—	**ADD R1, R2**
R1	5	**8**
R2	3	3

Decode reads ADD and its operands.
Execute adds 5 + 3 and writes 8 into R1.

Teaching model: 4-byte instructions, straight-line execution. The result table shows the full instruction.

Your turn: predict the result

Check yourself, question 1

ADD R5, R6 sits at address 0x300. R5 = 7, R6 = 2.

What are R5, R6 and the PC after the instruction executes?

R5 = ?	R6 = ?	PC = ?

Teaching model: 4-byte instruction, PC initially 0x300, straight-line execution. Predict before checking the answer.

The explained answer

Check yourself, question 1

ADD R5, R6 sits at address 0x300. R5 = 7, R6 = 2.

Fetch moves the PC on by one 4-byte instruction; execute writes the sum into the destination R5; the source R6 is only read.

R5 = 9	R6 = 2	PC = 0x304
7 + 2, written back	unchanged, read only	0x300 + 4

Source: Hardware/ANSWER-KEY.md, h03-05, Q1. The key explains the result for every register and the PC.

The same approach across three modules

Hardware

State
ADD R1, R2
PC 0x100, R1 = 5, R2 = 3

Transition
Fetch → Decode → Execute

Result
R1 = 8, R2 = 3
PC = 0x104

Networks

State
Fresh HTTPS over TCP
Configured host, no cached answer

Transition
DNS, routing, TCP, then TLS
HTTP inside TLS

Result
Validated server identity
Protected HTTP content in transit

Behaviorist AI

State
Q(A,X) = 20, reward = 10
B nonterminal, max Q(B,·) = 50

Transition
$\alpha = 1$, $\gamma = 0.9$
target = $10 + 0.9 \times 50 = 55$

Result
Qnew = $20 + 1 \times (55 - 20) = 55$
One table entry updates

Predict first. Check the explanation.

github.com/nsdmlk/bit-ai-and-cs Stage 2 due 7 November 2026